

The Mathematics of Multi Fractal Workflow: One Object, Every Level

A complete mathematical ascent — from counting and statistics to optimal transport, stratified geometry, multifractal spectra, and machine-checked foundations — organized around a single map $\tau : \mathcal{P}(\Pi) \rightarrow \mathcal{W}$

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Abstract

Multi Fractal Workflow (MFW) studies one map: the transformation τ that compiles PDDL 3.1 planning behavior into POWL v2 workflow structure. What makes MFW unusual is that this single object is simultaneously a statistics problem (a partition of a sample space), a combinatorics problem (counting linear extensions of partial orders), an algebra problem (a quotient by an equivalence relation), a logic problem (a machine-checked theory in dependent type theory), an information-theory problem (conditional entropy of a lossy code), a measure-theory and optimal-transport problem (geometry on conditional fiber measures), a differential-geometry problem (dimension loss on a stratified space), a functional-analysis problem (a vector space of invariant observables), a fractal-geometry problem (a bundle of Rényi dimensions), and an algebraic-topology problem (a simplicial complex of concurrency). This paper walks the entire ladder in order, rung by rung, defining every piece of mathematics as it is first needed and showing exactly where in MFW it lives. Every statement carries its verification standing from the Lean 4 formalization (`ProcInt.MFW`): `PROVEN`, `DEFINITION`, `CONJECTURAL`, or `BLOCKED`. Nothing is assumed beyond one college statistics course; nothing relevant is left out at the top.

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1 The one object

Everything in this paper is about a single map and its preimages:

$$\tau : (\mathcal{P}(\Pi), d_{\mathcal{P}}, \nu) \longrightarrow (\mathcal{W}, d_{\mathcal{W}}, \tau_*\nu).$$

Here $\mathcal{P}(\Pi)$ is the set of *lawful behaviors* of a PDDL 3.1 planning theory Π (timed action sequences that obey the physics, achieve the goal, and satisfy every temporal, numeric, and trajectory constraint), and $\mathcal{W}$ is the set of POWL v2 *workflow objects* (hierarchical models built from partial orders, which carry concurrency, and choice graphs, which carry decisions and cycles). The map τ is the compiler: behavior in, workflow class out. The decorations $d_{\mathcal{P}}, \nu, d_{\mathcal{W}}, \tau_*\nu$ — a metric, a measure, and their images — are exactly the higher mathematics this paper climbs toward.

The central open problem, stated once now and revisited at every rung:

Conjecture 1.1 (Crown theorem). [CONJECTURAL] $\tau(b_1) = \tau(b_2) \iff b_1 \sim_I b_2$, where $\sim_I$ is Mazurkiewicz trace equivalence under the action-independence relation I . Equivalently: $\ker(\tau) = \text{TraceEq}_I$.

Each level of mathematics below is a different pair of glasses for looking at this one statement and its consequences.

Ladder Note 1.2. The paper is a ladder. Each section states (a) the mathematics it introduces, at the level of a good first course in that subject; (b) the MFW objects built from it; (c) what is proven versus conjectured. Later sections freely use earlier ones.

2 Rung 0: Sets, functions, counting (the statistics you have)

2.1 Mathematics

A *set* is a collection; a *function* $f : X \rightarrow Y$ assigns each element of X one element of Y . f is *injective* if distinct inputs give distinct outputs, *surjective* if everything in Y is hit, *bijective* if both. A *partition* of X is a family of nonempty, pairwise-disjoint subsets covering X — exactly the “grouping variable” of statistics. The number of orderings of k distinct items is $k!$; the binomial coefficient $\binom{n}{2}$ counts pairs. A finite probability distribution on X assigns $p(x) \geq 0$ with $\sum_x p(x) = 1$.

2.2 In MFW

Definition 2.1 (Fiber). [DEFINITION] $F_w = \tau^{-1}(w) = \{b \in \mathcal{P}(\Pi) : \tau(b) = w\}$ — the group of behaviors the compiler merges into workflow w .

Theorem 2.2 (Fibers partition the sample space). [PROVEN] *Every b lies in $F_{\tau(b)}$ (*fiber-partition*), and distinct fibers are disjoint: $w_1 \neq w_2 \implies F_{w_1} \cap F_{w_2} = \emptyset$ (*fiber-disjoint*).*

Already at Rung 0 a design was falsified: a proposed workflow similarity used the Jaccard index $J(F_{w_1}, F_{w_2}) = |F_{w_1} \cap F_{w_2}| / |F_{w_1} \cup F_{w_2}|$. By disjointness $J \equiv 0$ off the diagonal, so the induced “metric” is the useless discrete one. A proven set-theory triviality killed a geometry proposal before implementation; the repair arrives at Rung 8.

Counting enters through fibers too:

Definition 2.3 (Fiber cardinality and probability). [DEFINITION] For a finite enumeration of behaviors, $|F_w|$ is a count, and $p(w) = |F_w| / |\mathcal{P}(\Pi)|$ is the fraction of behaviors landing in w — a genuine probability distribution on the image of τ ($p \geq 0$: [PROVEN]).

3 Rung 1: Relations, equivalence, and quotients (discrete algebra)

3.1 Mathematics

A *binary relation* on X is a set of pairs. An *equivalence relation* $\equiv$ is reflexive ($x \equiv x$), symmetric, and transitive; its *equivalence classes* $[x] = \{y : y \equiv x\}$ form a partition, and conversely every partition arises this way. The *quotient* $X/\equiv$ is the set of classes. The *kernel* of a function f is the equivalence $x \equiv_f y \iff f(x) = f(y)$; the first isomorphism idea: f factors as $X \twoheadrightarrow X/\equiv_f \hookrightarrow Y$. A *strict partial order* $\prec$ is irreflexive and transitive; a *linear extension* of $\prec$ on n items is a total order compatible with it, formally a bijection σ with $i \prec j \implies \sigma(i) < \sigma(j)$.

3.2 In MFW

The kernel of the compiler is the star of the show:

Definition 3.1 (KernelEquiv). [DEFINITION] $b_1 \equiv_K b_2 \iff \tau(b_1) = \tau(b_2)$. Equivalence laws: [PROVEN].

Independence and traces give the *other* equivalence:

Definition 3.2 (Independence; Mazurkiewicz trace equivalence). [DEFINITION] $I \subseteq A \times A$ is symmetric; $(a, b) \in I$ demands five commutation facts (effects commute; preconditions stable; invariants stable; numeric flows compatible; trajectory constraints preserved). Trace equivalence $u \sim_I v$ on action sequences is the reflexive-symmetric-transitive closure of the adjacent swap $uabv \rightsquigarrow ubav$ for $(a, b) \in I$. That $\sim_I$ is an equivalence relation: [PROVEN] (traceEquivEquivalence).

The crown conjecture (1.1) now reads as pure Rung-1 algebra: *two independently defined equivalence relations on the same set are claimed equal*. The formalization proves the claim is non-circular ([PROVEN]): $\equiv_K$ unfolds using only τ and equality in $\mathcal{W}$; $\sim_I$ unfolds using only I and event lists — neither side mentions the other’s vocabulary (kernelEquivUnfoldsWithoutTraces, traceEquivUnfoldsWithoutTau, both rfl).

Partial orders enter as the *content* of a trace class:

Definition 3.3 (Causal order; trace class). [DEFINITION] Each behavior b under I induces a causal order P_b ($i \prec j$ iff dependent and earlier; irreflexive, transitive). The trace class is $[b]_I = \{b' : b \sim_I b'\}$.

Theorem 3.4 (Fiber = trace class, conditionally). [PROVEN] *under hypothesis* If the crown holds (taken as an explicit hypothesis), then $F_{\tau(b)} = [b]_I$ as sets (fiberEqTraceClass).

Conjecture 3.5 (Trace class $\simeq$ linear extensions). [CONJECTURAL] *Under bijection hypotheses (distinct events, decidable I , independence exactly complementing transition-level dependence): $[b]_I \simeq \text{Lin}(P_b)$. Classical (Mazurkiewicz 1977; Diekert–Rozenberg 1995) but not yet formalized under PDDL 3.1 semantics.*

There is also an *order on the transformations themselves*: $\tau_1 \sqsubseteq \tau_2$ (“refines”) iff τ_1 -equality implies τ_2 -equality, i.e. $\tau_2 = \psi \circ \tau_1$; this is reflexive and transitive ([PROVEN]) and will matter at Rung 6.

4 Rung 2: Graphs, Petri nets, and structural decomposition

4.1 Mathematics

A *directed graph* is nodes plus arrows. A *Petri net* $N = (P, T, F)$ is a bipartite directed graph of *places* and *transitions*; tokens on places move by *firing* transitions (consume one token per input place, produce one per output place). A *workflow net* has a unique source place, unique sink place, and every node on a path between them; it is *safe* if no place ever holds two tokens and *sound* if there are no dead transitions, every reachable state can still finish, and completion is clean. Two structural subclasses: *state machines* (every transition: ≤ 1 input, ≤ 1 output place — all branching is choice) and *marked graphs* (every place: ≤ 1 input, ≤ 1 output transition — all branching is concurrency).

4.2 In MFW and its companion conversion theory

POWL v2 is defined recursively (Kourani–van Zelst):

Definition 4.1 (Powl model, well-formedness). [DEFINITION]

$$\text{Powl}(\alpha) ::= \text{atom}(a) \mid \text{silent} \mid \text{xor}(\vec{c}) \mid \text{loop}(c_1, c_2) \mid \text{po}(\vec{c}, \prec),$$

with hereditary well-formedness ($|\vec{c}| \geq 2$ for xor; $\prec$ irreflexive and transitive for po). POWL v2 adds *choice graphs* (directed start-to-end graphs over submodels, expressing non-block decisions and cycles); a POWL v2 object carries the model, its well-formedness proof, hierarchical depth, boundary variables ∂w , and choice graphs.

The structural duality — partial order $\leftrightarrow$ marked graph, choice graph $\leftrightarrow$ state machine — powers the proven conversion theory for the companion question (Kourani–Park–van der Aalst 2026): *separable* WF-nets (recursively nested state machines and marked graphs, never cross-linked) convert to POWL 2.0 *completely* (language equality, their Thm. 5.5) and *totally on the class* (their Thm. 5.6); explicit sound free-choice counterexamples (long-term dependencies; loops entering one parallel branch; interleaved choice/concurrency at a shared synchronization) bound the class from outside. In MFW’s terms this settles the *existence* and *exactness* legs of the compilation question; the *maximality* leg is the crown conjecture.

5 Rung 3: Combinatorics of order — counting concurrency

5.1 Mathematics

For a finite poset P , $e(P) = |\text{Lin}(P)|$ counts linear extensions — between 1 (a chain) and $n!$ (an antichain). An *antichain* is a set of pairwise-incomparable elements; the *width* $w(P)$ is the largest antichain, and Dilworth’s theorem says $w(P)$ equals the minimum number of chains covering P . *Interval orders*: if each event occupies a real interval $[s_a, e_a]$, then $a \prec b \iff e_a < s_b$.

5.2 In MFW

Definition 5.1 (Serialization entropy). [DEFINITION] $H_{\text{ser}}(P) = \log e(P)$: the information content of accidental serialization. One concurrency class of k mutually independent actions carries $H_{\text{ser}} = \log k!$ — the compression opportunity of the transformation is *factorial*.

Definition 5.2 (Erasure target). [DEFINITION] The least causal relation preserving correctness: $R^* = \arg \min_R |R|$ subject to $\text{Lin}(E, R) \subseteq \text{Valid}(\Pi)$. “Maximize concurrency” means emit R^* .

Definition 5.3 (Width; temporal abstraction). [DEFINITION] $w(P)$ is the primary concurrency coordinate (one scalar — deliberately too weak; its multiscale refinement is Rung 9). The abstraction ladder $\text{MetricTime} \rightarrow \text{IntervalTime} \rightarrow \text{CausalTime} \rightarrow \text{HierarchicalTime}$ erases timestamps in stages; interval orders sit in the middle. The governing slogan, recorded in the source: *the question determines the sufficient abstraction*.

Definition 5.4 (Temporal restriction gap). [DEFINITION] With a simple temporal network N (difference bounds $\ell_{ij} \leq t_j - t_i \leq u_{ij}$), $\text{Lin}_T(P, N)$ is the temporally feasible extensions, $H_T(P, N) = \log |\text{Lin}_T(P, N)|$ (opaque interface in Lean), and $H_{\text{ser}}(P) - H_T(P, N) \geq 0$ measures causal concurrency the clock forbids. Four concurrency types are kept distinct: $C_E = C_C \cap C_T \cap C_R$ (executable = causal $\cap$ temporal $\cap$ resource).

6 Rung 4: Logic and type theory — the machine-checked substrate

6.1 Mathematics

In *dependent type theory* (the logic of Lean 4), propositions are types and proofs are their inhabitants. A *subtype* $\{x : X \mid P(x)\}$ packages an element with a proof, so “being lawful” can be made impossible to fake rather than merely asserted. Definitional equality (**rf1**) distinguishes unfoldings from theorems. *Decidability* of a relation is itself a mathematical asset (it licenses computation inside proofs).

6.2 In MFW

This rung explains several audit repairs that are really logic:

- **Proof-carrying behaviors.** The old `LawfulBehavior` stored lawfulness as data fields, so a behavior with `achievesGoal := False` inhabited the “lawful” space. The repair is the subtype discipline: `BehaviorTrace` (raw data) + `IsLawful` (predicate, five clauses) + `LawfulBehavior` (the pair). *The proof is a witness, not a payload.*
- **Semantics or absence.** A constraint type without a satisfaction relation does not constrain anything; hence Π carries $\models_T, \models_N, \models_\Gamma$ explicitly.
- **No vacuous laws.** A former field `respects_equivalence` concluded `True` from an assumption already containing its conclusion; it was deleted, and the kernel characterization became a *theorem target* instead of an assumption — the difference between logic and wishful typing.
- **Non-circularity as rf1.** Both crown sides unfold definitionally in disjoint vocabularies; the biconditional is therefore contentful.
- **Standing discipline.** Every declaration carries `PROVEN / DEFINITION / CONJECTURAL / BLOCKED`; this paper inherits that labeling.

7 Rung 5: Information theory — what the compiler erases

7.1 Mathematics

Shannon entropy of a uniform distribution on m items is $\log m$; conditional entropy $H(X | Y) = \sum_y p(y) H(X | Y = y)$ measures what remains unknown about X after seeing Y ; conditional mutual information $I(X; Y | Z) \geq 0$ measures shared information given Z . A statistic T is *sufficient* for Y when $I(Y; X | T) = 0$ (Fisher–Neyman).

7.2 In MFW

The compiler is a lossy code; the loss is exactly quantified:

Definition 7.1 (Fiber entropy; total fiber entropy). [DEFINITION]

$$S_\tau(w) = \log |F_w|, \quad H(B | W) = \sum_w p(w) S_\tau(w), \quad 0 \leq H(B | W) \leq \log |\mathcal{P}(\Pi)|.$$

Nonnegativity, the zero-iff-singleton characterization ($S_\tau(w) = 0 \iff |F_w| \leq 1$; for reachable w , injectivity), monotonicity in fiber size, the total-collapse identity ($S = \log |\mathcal{P}(\Pi)|$ when everything maps to one class), and the upper bound all have complete Lean proof terms but carry in-source [CONJECTURAL] labels pending API review; congruence and $|F_w| > 0$ for reachable w are [PROVEN].

Remark 7.2 (Additivity repair). $\log |F_w|$ is not additive over disjoint fibers; the additive atom is $\eta(w) = p(w) \log |F_w|$, and $H(B | W) = \sum_w \eta(w)$ distributes correctly over hierarchies. This became the *seventh* measure kind after an audit found fiber entropy conflated with temporal slack — they are orthogonal: $\text{Slack}(w) \neq H(B | W = w)$.

Conjecture 7.3 (Entropy collapse). [CONJECTURAL] *If the crown and the trace-class bijection both close:*

$$S_\tau(\tau(b)) = \log |F_{\tau(b)}| = \log |[b]_I| = \log |\text{Lin}(P_b)| = \log e(P_b) = H_{\text{ser}}(P_b).$$

Fiber entropy equals serialization entropy: an information-theoretic quantity (Rung 5) collapses onto a combinatorial one (Rung 3), and the erased information becomes computable from the emitted diagram alone.

Sufficiency completes the rung: τ is *sufficient* for a target property Y when Y factors through τ (deterministic fragment, [DEFINITION]); the stochastic bridge $I(Y; B | W) = 0 \implies$ sufficiency is [CONJECTURAL] (placeholder structure `SemanticResidualInfo`, awaiting measure-theoretic conditional entropy — Rung 8’s machinery).

8 Rung 6: Set-level functional structure — observability and sigma-algebras

8.1 Mathematics

A *sigma-algebra* on X is a family of subsets closed under complement and countable union; the sigma-algebra *generated by* a map f is $\{f^{-1}(U)\}$ — the events decidable by observing f only. A *Boolean algebra* of propositions is closed under $\neg, \wedge, \vee, \rightarrow$. A *vector space* of functions is closed under pointwise $+$ and scalar $\cdot$; a subset *spans* if finite linear combinations reach everything; a family is a *basis* if it spans and is linearly independent.

8.2 In MFW

Definition 8.1 (Observable / hidden / fiber-constant). [DEFINITION] $P : \mathcal{P}(\Pi) \rightarrow \text{Prop}$ is *observable* iff it factors: $\exists \hat{P}, P = \hat{P} \circ \tau$; *fiber-constant* iff $\tau(b_1) = \tau(b_2) \implies (P(b_1) \leftrightarrow P(b_2))$; *hidden* iff some fiber contains a P -witness and a $\neg P$ -witness.

Theorem 8.2 (The characterization and its algebra). *Observable $\iff$ fiber-constant (forward [PROVEN]; reverse constructed via the factoring predicate, labeled [CONJECTURAL] in-source though the term closes). Hidden $\iff$ not fiber-constant; observable $\implies$ not hidden ([PROVEN]). Observables form a Boolean algebra containing $\top, \perp$ and all pullbacks, closed under every unary and binary connective ([PROVEN]).*

Definition 8.3 (Observable sigma-algebra; information horizon). [DEFINITION] $\mathcal{O}_\tau = \{S \subseteq \mathcal{P}(\Pi) : S \text{ saturated}\} = \{\tau^{-1}(U) : U \subseteq \mathcal{W}\}$ — both inclusions [PROVEN]. The *information horizon* H_τ is the observable/hidden dichotomy; inside it, everything is determinable from the POWL object; outside, erased. Refinement $\tau_1 \sqsubseteq \tau_2$ monotonically enlarges the observable algebra ([PROVEN]) — the principled dial for “make Y observable, and no more.”

Definition 8.4 (Invariant observables F_τ). [DEFINITION] $F_\tau = \{f : \mathcal{P}(\Pi) \rightarrow \mathbb{R} \mid f \text{ constant on fibers}\}$. It contains 0 and constants; closed under $+$, scalar, negation, and product ([PROVEN]); every $g \circ \tau$ belongs ([PROVEN]); conversely every $f \in F_\tau$ factors through surjective τ by choosing fiber representatives (term closes; [CONJECTURAL] label in-source). Hence morally $F_\tau \cong (\mathcal{W} \rightarrow \mathbb{R})$; for finite $\mathcal{W}$, $\dim F_\tau = |\text{im } \tau|$. This is the collider-physics analogy made exact: F_τ plays the role of infrared/collinear-safe observables, blind to what the transformation erases.

Definition 8.5 (Basis program; truncation). [DEFINITION] Five primitive kinds (causal incidence, interval, motif, hierarchical, temporal-boundary — EFP analogues: angular structure, energy flow, graph polynomials, jet substructure, thrust). $\text{Span} \subseteq F_\tau$: complete induction proof (labeled [CONJECTURAL]). The basis claim — $\text{span}\{\varphi_i\} = F_\tau^{\text{adm}}$ under an admission profile — is [CONJECTURAL], with conjectured truncation bounds $\#\text{causal} \leq \binom{\max_{\text{Conc}}}{2}$, $\#\text{motif} \leq 2^{\binom{\max_{\text{Branch}}}{2}}$, total order polynomial in the profile. Feature extraction $b \mapsto [\varphi_1(b), \dots, \varphi_n(b)]$ is fiber-constant ([PROVEN]): equivalent behaviors get identical feature vectors.

9 Rung 7: Probability refined — the q-lens and Rényi reweighting

9.1 Mathematics

Given a positive finite distribution p , the *escort* (or Rényi/thermodynamic) deformation at inverse-temperature-like parameter $q \in \mathbb{R}$ is $p^{(q)}(x) = p(x)^q / \sum_y p(y)^q$. As q grows it concentrates on the mode; $q = 1$ returns p ; $q = 0$ is uniform on the support; $q < 0$ inverts attention onto rare outcomes.

9.2 In MFW

Theorem 9.1 (q-lens). [PROVEN] $L_q(p)$ is a well-defined distribution (positivity, normalization), and the ratio law holds:

$$\frac{L_q(p)(x)}{L_q(p)(y)} = \left(\frac{p(x)}{p(y)} \right)^q \quad (\text{qLens_ratio_law}).$$

Interpretation fixed by the source: q is *attention control over behavioral mass inside POWL choice structure* — $q \gg 0$ audits dominant channels, $q \ll 0$ audits the rare-behavior tail, $q = 1$ is the Shannon scale. The q -lens also parameterizes *search rails* (Rung 10’s portfolio) and, at Rung 9, the multifractal spectrum.

10 Rung 8: Measure theory and optimal transport — geometry on workflows

10.1 Mathematics

A *measure* ν assigns sizes to sets; the *pushforward* $\tau_*\nu$ transports it along a map ($\tau_*\nu(U) = \nu(\tau^{-1}U)$); a *conditional measure* $\nu_w = \nu(\cdot \mid \tau = w)$ restricts and renormalizes to a fiber. A *metric* satisfies identity, symmetry, triangle inequality (*pseudo*-metric: distinct points may sit at distance 0). Given a ground metric d , the *Wasserstein distance* between distributions is the optimal-transport cost $W_p(\mu, \rho) = \inf_{\pi \in \Pi(\mu, \rho)} \mathbb{E}_\pi[d(x, y)]$, infimum over couplings with the given marginals (Villani).

10.2 In MFW

This is the repair of the Rung-0 Jaccard collapse:

Definition 10.1 (Wasserstein workflow metric). [DEFINITION] Fix a behavior metric $d_{\mathcal{P}}$ on $\mathcal{P}(\Pi)$ (axioms carried as proof fields). Define

$$d_{\mathcal{W}}(w_1, w_2) = W_p(\nu_{w_1}, \nu_{w_2}).$$

Fibers are disjoint as *sets*, but their behavior *distributions* can be close in $d_{\mathcal{P}}$ — distance lives at the level of measures, not supports. In Lean the distance is **opaque** (interface; the transport construction needs measure theory beyond the module). A **WorkflowMetric** record induces a **PseudoMetricSpace** ([CONJECTURAL] pending the distance’s own well-definedness).

Definition 10.2 (Pushforward masses; provenance law). [DEFINITION] **PushforwardMass**: $\mu : \mathcal{W} \rightarrow \mathbb{R}_{\geq 0}$. The full *vector measure* has seven components $[\mu_B, \mu_T, \mu_C, \mu_L, \mu_{Sl}, \mu_F, \mu_S]$ (behavioral, temporal, choice, linearization, slack, fluent, entropic), with the standing *provenance requirement*: each must ultimately be derived from (τ, ν) , not posited.

Rung 8 also supplies what Rung 5 lacked: measure-theoretic conditional entropy $H(B \mid W = w)$ generalizing $\log |F_w|$ to infinite fibers, and the machinery for the sufficiency bridge $I(Y; B \mid W) = 0$ — both explicitly deferred as research targets in the source.

11 Rung 9: Fractal geometry — the multifractal spectrum bundle

11.1 Mathematics

For a measure spread over a nested family of partitions at scales k , the *partition sum* $Z(q, k) = \sum_{B \in P_k} p_B^q$ generates the *Rényi (generalized) dimensions* D_q from its scaling in k ; D_0 is box-counting (capacity), D_1 the information dimension, D_2 correlation. A measure is *multifractal* when D_q genuinely varies with q — different moments see different scaling. Z is classically log-convex in q .

11.2 In MFW

POWL v2 supplies the canonical multiscale filtration for free: the hierarchy $P_0 \preceq P_1 \preceq \dots \preceq P_h$ of the workflow object (refinement law currently concludes **True** — the is-submodel-of predicate is [\[CONJECTURAL\]](#)).

Definition 11.1 (Spectrum bundle). [\[DEFINITION\]](#)

$$\vec{D}_q = (D_q^B, D_q^T, D_q^C, D_q^L, D_q^{Sl}, D_q^F),$$

one generalized dimension per measure kind: a section of a trivial $\mathbb{R}^6$ -bundle over the q -line, with the axiom that D_0 is finite for each kind. *Provenance (audit)*: the bundle must be *manufactured* from admitted measure + scale + fit evidence, never constructed as an arbitrary function — without the kernel, “spectrum” is a name, not a derived object.

Definition 11.2 (Cross-spectrum tension). $\Delta_{ij}(q) = D_q^i - D_q^j$. Antisymmetry, diagonal vanishing, zero trace: [\[PROVEN\]](#). The interpretation dictionary: $D_q^B \gg D_q^T$ temporal bottleneck; $D_q^B \ll D_q^T$ scheduling elasticity; $D_q^C \gg D_q^B$ choice sparsity; $D_q^L \gg D_q^T$ *false concurrency* — Rung 3’s temporal restriction gap reappearing as a spectral signature. The source is explicit: *the optimization signal is the tension profile, not any single spectrum*.

Definition 11.3 (Partition sums; scale resolution). $Z(q, k) = \sum_{B \in P_k} p_B^q$ with $Z(0, k) = |P_k|$ (term closes, requires positivity; [\[CONJECTURAL\]](#) label), $Z(1, k) = 1$ under normalization ([\[PROVEN\]](#)), nonnegativity and disjoint additivity (terms close; [\[CONJECTURAL\]](#) labels). The scale-resolved bundle $\vec{D}_q(k)$ adds hierarchical depth; tension antisymmetry persists at every depth ([\[PROVEN\]](#)). Open: log-convexity of Z in q ; whether the bundle is a genuine fiber bundle with interesting structure group; a Fisher-metric reading of the profile space.

Definition 11.4 (Transformation information profile). [\[DEFINITION\]](#) The master diagnostic $I_\tau = (d_{\text{int}}, \vec{D}_q, H(B \mid W), I(Y; B \mid W), \dim F_w, d_W, \Delta_{ij}(q))$ — one record consolidating Rungs 5–9; the cached tension provably matches the bundle ([\[PROVEN\]](#)).

Remark 11.5 (Why “multi fractal workflow”). The name is now redeemable in full: the pushforward of behavioral measure onto the POWL hierarchy is (conjecturally) multifractal — different semantic freedoms (behavior, time, choice, linearization, slack, fluents) scale with genuinely different exponents across the same hierarchy, and the workflow’s structure is read off from the *spread* of $\vec{D}_q$, not from any single dimension.

12 Rung 10: Stratified differential geometry — dimension loss

12.1 Mathematics

A *manifold* looks locally like $\mathbb{R}^d$; its *dimension* is d . Spaces cut out by equalities $C(x) = 0$ and inequalities $G(x) \leq 0$ with discrete choices are not manifolds but *stratified spaces*: unions of manifolds (*strata*) of varying dimension, glued along boundaries. A *submersion* has surjective differential; the preimage/fiber theorem then makes fibers submanifolds of dimension $\dim(\text{source}) - \dim(\text{target})$. Lipschitz maps cannot raise dimension. The *tangent space* at a point collects infinitesimal directions; a *null space* of a map collects directions it does not see.

12.2 In MFW

Definition 12.1 (Stratification of the phase space). [DEFINITION] Coordinates come in five families $x = (\text{event occurrence, temporal, object-fluent, numeric, trajectory})$; discrete choice patterns $\sigma \in \Sigma$ induce $\mathcal{P}(\Pi) = \bigcup_{\sigma} P_{\sigma}$ with covering law, each stratum carrying a local dimension (continuous freedom: slack, numeric state).

Definition 12.2 (Dimension loss). [DEFINITION] $\Delta d_{\tau}(b) = d_{\text{loc}}^{\text{PDDL}}(b) - d_{\text{loc}}^{\text{POWL}}(\tau(b)) \in \mathbb{Z}$. Strictly stronger information than cardinality: an uncountable fiber (continuous slack) can be low-dimensional; a finite fiber has dimension 0. Honest ledger: the POWL-side local dimension is currently the placeholder 0 (untimed structural POWL), so the nonnegativity “proof” is vacuously easy; the real statement needs a Lipschitz hypothesis ([CONJECTURAL]). “ $\Delta d_{\tau} = 0 \iff$ local invertibility” and “fiber dimension = dimension loss under submersion” are [BLOCKED]: they require smooth structure on $\mathcal{P}(\Pi)$ and $\mathcal{W}$ not yet formalized.

Definition 12.3 (Null perturbation space). [DEFINITION] $\text{Null}_{\tau}(b) = \{\delta : \tau(b \oplus \delta) = \tau(b)\}$, the lawful perturbations invisible to the compiler — the fiber tangent space at regular points. Proven basics: the trivial perturbation is null; kind-restricted null spaces are subsets; null perturbations preserve $\sim_{\tau}$ and stay in the fiber ([PROVEN]). The kind decomposition $\Delta d_{\tau} = \Delta d^{\text{temp}} + \Delta d^{\text{fluent}} + \Delta d^{\text{num}} + \Delta d^{\text{traj}}$ is [BLOCKED] (needs direct-sum tangent decomposition); per-kind dimensions are placeholders. The measure link: if the source measure is d -dimensional and fibers are k -dimensional, $\mu(B_{\varepsilon}(w)) \sim \varepsilon^{d-k} \text{Vol}_k(F_w \cap B_{\varepsilon})$ — dimension loss controls how Rung 8’s pushforward concentrates, feeding Rung 9’s exponents.

The physics-flavored open question of this rung, imported from black-hole thermodynamics via the fiber-entropy scaling models (Rung 5 $\times$ Rung 10): does $S_{\tau}(C)$ scale with component *volume* $|C|$ or *boundary* $|\partial C|$ (*boundary sufficiency hypothesis*, $\exists \beta > 0 : |S_{\tau}(C) - \beta|\partial C|| \leq \varepsilon$)? An area law would mean the POWL interface variables suffice to reconstruct interior behavioral information — a holographic property of the compiler. [DEFINITION] (hypotheses stated; empirical status open).

13 Rung 11: Algebraic topology — the shape of concurrency

13.1 Mathematics

An *abstract simplicial complex* is a family of finite sets closed under subsets — a combinatorial space whose k -faces are $(k+1)$ -element sets. *Simplicial homology* H_k counts k -dimensional holes; *persistent homology* tracks holes across a filtration parameter. A complex is *contractible* when it has no holes at all.

13.2 In MFW

Definition 13.1 (Concurrency complex). [DEFINITION] $K_{\Pi} = \{A \subseteq E : A \text{ jointly concurrent}\}$. Downward closure and $\emptyset \in K$ are *admission hypotheses*, not free (the audit removed sorry’d closure proofs — they are false for arbitrary predicates, e.g. “ $|A| = 2$ ”); with them, K_{Π} is a genuine simplicial complex with dimension $\max_A |A| - 1$ ([PROVEN] construction given the hypotheses).

The research program: $H_k(K_\Pi)$ and its persistence as temporal or constraint thresholds vary would measure the *shape* of concurrency (not just its width); the transformation $\text{PDDL} \rightarrow K_\Pi \rightarrow \text{POWL}$ may be selecting a hierarchical representation of this complex. *Falsifier recorded in-source*: the thesis dies if K_Π is trivially contractible for all admitted planning theories. All homological claims: [\[CONJECTURAL\]](#).

14 Rung 12: The generator picture — presentation of the kernel

14.1 Mathematics

An equivalence relation can be *presented* by generators: primitive moves whose reflexive-symmetric-transitive closure it is (as groups are presented by generators and relations). If two generator types are independent (neither’s moves are expressible by the other’s), quantities counting them are independent coordinates.

14.2 In MFW

Definition 14.1 (Kernel generators and paths). [\[DEFINITION\]](#) $g \in \{\text{traceSwap}, \text{choiceNormalization}, \text{hierarchyCo}\}$ a *kernel path* $b_1 \rightarrow^{g_1} \dots \rightarrow^{g_n} b_2$ carries generators, intermediates, and endpoint proofs; $\kappa(\text{path})$ counts each type.

Conjecture 14.2 (Kernel generated). [\[CONJECTURAL\]](#) $b_1 \equiv_K b_2 \iff \exists \text{ kernel path.}$ If it holds, the generator types determine which erasure modes are independent, hence which spectra exist: one D_q per independent generator — the proof-first derivation of Rung 9’s basis, replacing enumeration of measure kinds by presentation of the kernel. This is the deepest expression of the paper’s organizing principle: first characterize what the map forgets; all structure follows.

15 Rung 13: Governance mathematics — the control plane

The remaining formal modules are interface-level specifications (opaque carriers, Prop-valued laws) that wrap the analytic core in an engineering discipline; their mathematics is modest but their *shape* is load-bearing:

- **Manufacture / BRCE** (§§1–6, 15): manufacturing law $\mathcal{L} \subseteq \mathcal{A} \times \mathcal{O}$, receipts, and the invariant $\forall a : \text{isReceipted}(a)$ (zero unreceipted actuation); Rice containment scopes verification to an admitted semantic domain — a nod to Rice’s theorem that arbitrary semantic properties are undecidable. [\[DEFINITION\]](#)
- **Falsification** (§38): status $\{\text{Admitted}, \text{Revoked}\}$; $\text{RevCond} = (\text{Admitted} \wedge F(T, E))$; $\text{FalsLoop} = (\text{RevCond} \implies \text{status}(\text{revoke}) = \text{Revoked})$ — Popper as an interface. Composition proven from hypotheses in tests. [\[DEFINITION\]](#)
- **Explore/Exploit** (§§7–12, 35): exploitation factors through contracts (a congruence-shaped separation law). [\[DEFINITION\]](#)
- **Search portfolio** (§§31–35): an exact completeness rail F_0 plus q-lens-parameterized exploitation rails; conjecture: completeness + persistent service + noninterference $\implies$ portfolio completeness. [\[CONJECTURAL\]](#)
- **Compiler pipeline** (§§36–37): the cascade $\text{Coords} \rightarrow \text{TTL} \rightarrow \text{Tera} \rightarrow \text{RustExe}$ with stage equalities carried as proof fields and the bound $\text{multDepth} \leq \text{time}$. [\[DEFINITION\]](#)

- **Ledger** (§§39–40): claims as a DAG with derivations in topological order — *known stub*: `validTopologicalSort := True`; the check validates nothing yet. [DEFINITION] (placeholder)

16 The whole ladder at a glance

Rung	Mathematics	MFW object	Peak standing
0	sets, partitions, counting	fibers; $p(w)$	[PROVEN] (disjointness)
1	equivalence, quotients, posets	$\equiv_K, \sim_I, [b]_I$, crown	[CONJECTURAL] (crown)
2	graphs, Petri/WF-nets	POWL v2; separable conversion	[PROVEN] (Thms 5.5/5.6)
3	linear extensions, Dilworth	H_{ser}, R^*, H_T gap, C_E	[DEFINITION]
4	dependent type theory	proof-carrying behaviors; rfl	[PROVEN] (non-circularity)
5	Shannon entropy, sufficiency	$S_\tau, H(B W), \eta$, collapse	[CONJECTURAL] (collapse)
6	sigma-algebras, vector spaces	$\mathcal{O}_\tau, F_\tau$, basis	[PROVEN] (algebra of F_τ)
7	escort distributions	q-lens, ratio law	[PROVEN]
8	measures, optimal transport	$\tau_*\nu, \nu_w, d_{\mathcal{W}} = W_p$	[DEFINITION] (opaque)
9	Rényi dimensions, multifractals	$\vec{D}_q, \Delta_{ij}(q), Z(q, k)$	[PROVEN] (tension algebra)
10	stratified geometry	$\Delta d_\tau, \text{Null}_\tau$, area law	[BLOCKED] (smoothness)
11	simplicial complexes, homology	K_Π , persistence	[CONJECTURAL]
12	presentations by generators	kernel paths, spectrum basis	[CONJECTURAL]
13	specification logic	BRCE, falsification, ledger	[DEFINITION]

Reading the table column-wise gives the paper’s three theses. First (column 2): no rung is decorative — each field enters because a specific question about τ cannot be asked without it. Second (column 3): every object at every level is derived from, or aimed at, the kernel; the crown conjecture is the single point of failure and the single point of unification. Third (column 4): the formalization keeps an exact ledger of what is settled. The proven layer (partitions, trace algebra, conversion theorems, non-circularity, observable algebra, q-lens, tension algebra) is already enough to build and audit a compiler; the conjectural layer (crown, collapse, basis, generation) says what such a compiler would *mean*; the blocked layer (smooth structure) marks where new formalized mathematics — not new ideas — is the bottleneck.

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